

Slug ascent at Stromboli.

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1 Method

The ascent of a gas slug at Stromboli is modeled using MagmaFOAM. Ground deformation signals generated by slug ascent are calculated using syntheticSignals, and compared to seismic and strain records in the VLP band (Chouet et al., 2003; Mattia et al., 2021).

Pressure at the boundaries of the fluid system is used to calculate ground displacement at the Earth’s surface (Longo et al., 2012).

2 Slug ascent

The conduit is 170 m long and 30 m wide. Gas enters from the bottom, forming Taylor bubbles that rise towards the top (movie).

3 Ground deformation

References

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